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Harnessing Generative AI Using OpenAI APIs





Prerequisites

- Basic Python programming
- An account with <https://platform.openai.com/>
- (Paid account, can top up for \$5)
- Course content: <https://github.com/janani-ravi-loony/hands-on-openai-apis>
- Course content geared towards a technical audience
 - Developers, testers, analysts, program and project managers - anyone who works in technology



Generative AI

Generative Artificial Intelligence (Generative AI) is AI capable of generating text, images, or other media using generative models



Generative AI

- Powered by very large models called foundation models that are trained on huge datasets
- Can perform out-of-the-box tasks such as question answering, summarization, classification





How Do Generative AI Models Work?

- Powerful ML model to learn patterns and relationships in a dataset created by humans
- Model uses learned data to create new content
- New content resembles the content that the model has already seen before



OpenAI



- Organization focusing on artificial intelligence research and development
- Made significant contributions to the field of AI developing influential models such as GPT-3, GPT-4, GPT-4o, GPT-5





ChatGPT



Language model developed by OpenAI. Designed for natural language understanding and generation, capable of performing a wide range of text-based tasks

Large Language Models (LLMs)



- These are models that process natural language inputs and **predicts the next word based** on what has come before
- They are large because the models themselves are huge – they have billions of parameters that need to be trained
- They are also large because they are trained on a very large corpus of data comprising of billions of records





Attention-based Models and Transformers

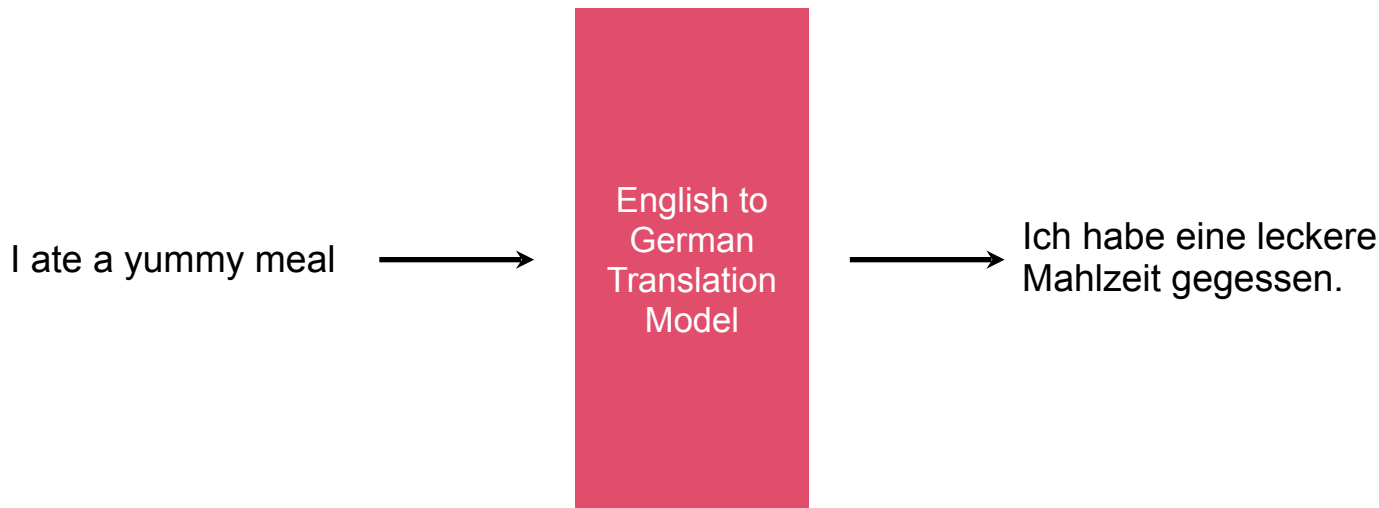


Attention in Neural Networks

Mechanism that allows the network to focus on specific parts of the input data while ignoring others



Language Translation Model





Model Without Attention

I ate a yummy meal



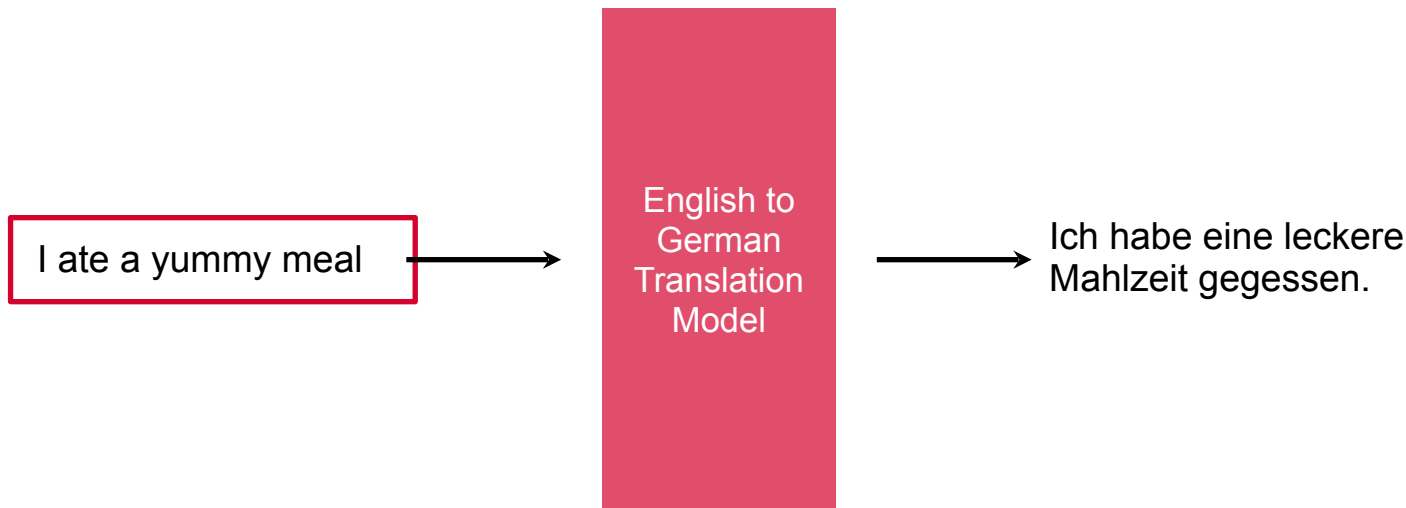
English to
German
Translation
Model



Ich habe eine leckere
Mahlzeit gegessen.

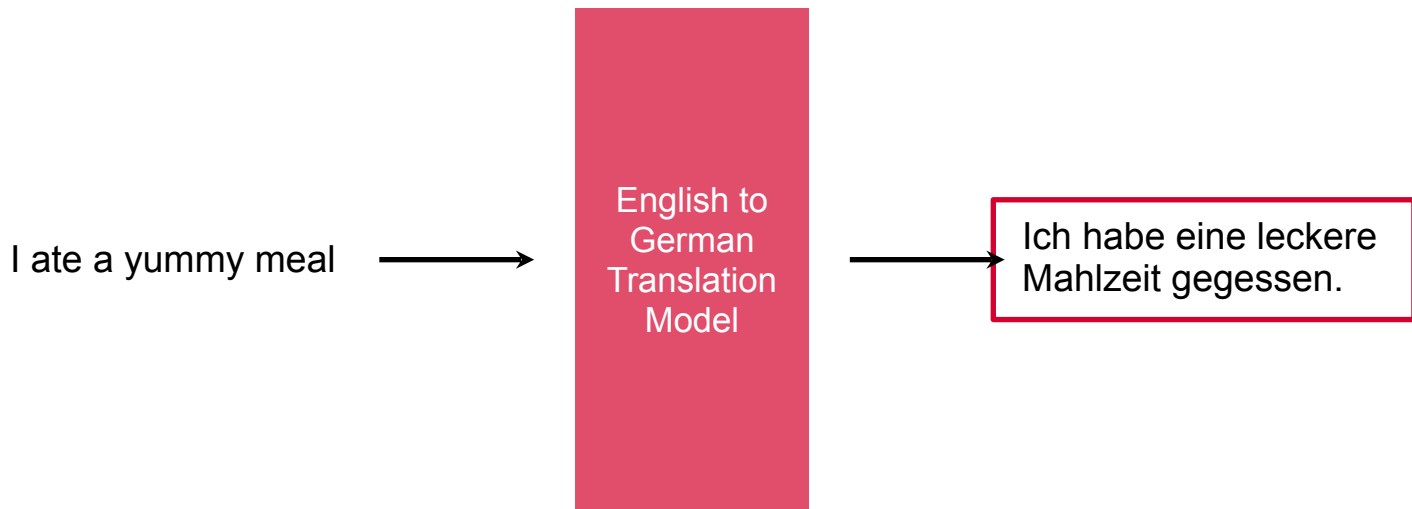


Will Parse the Entire Input

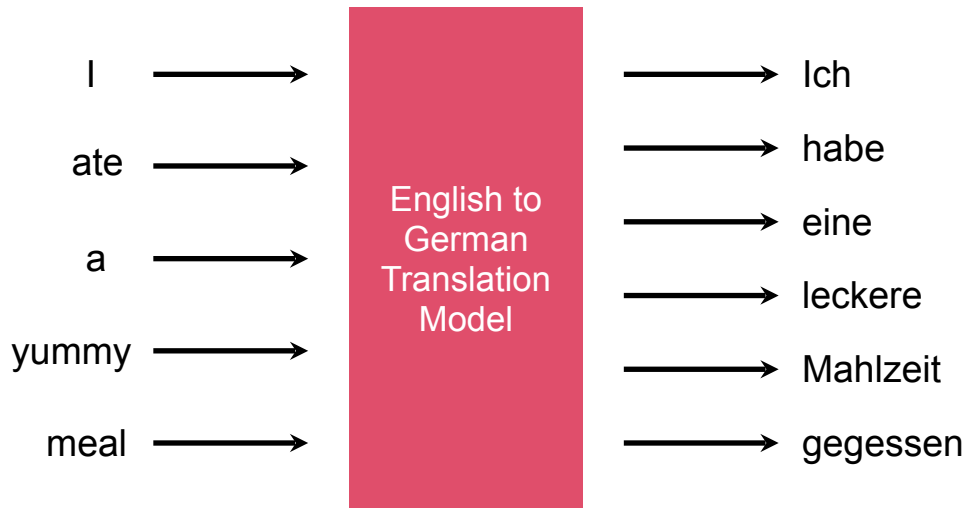




And Produce the Translation

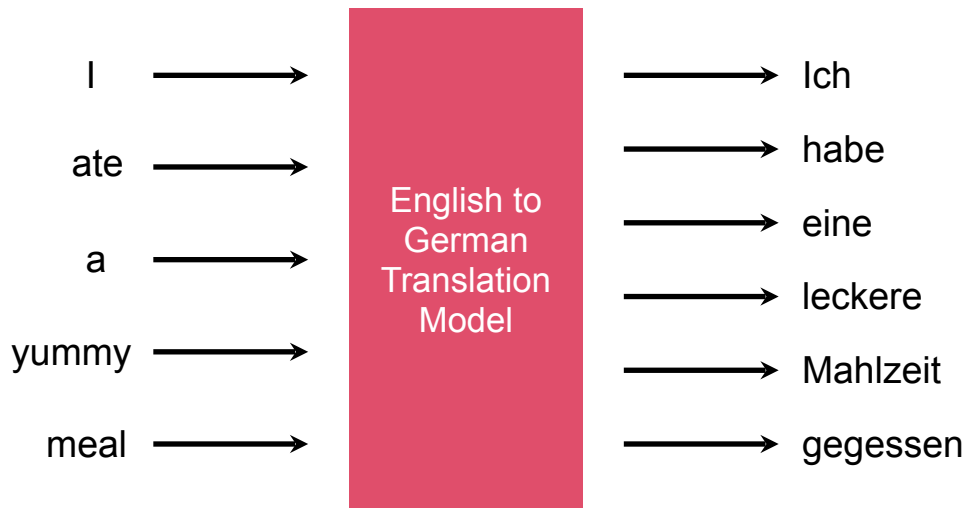


Words Fed and Output as Sequences



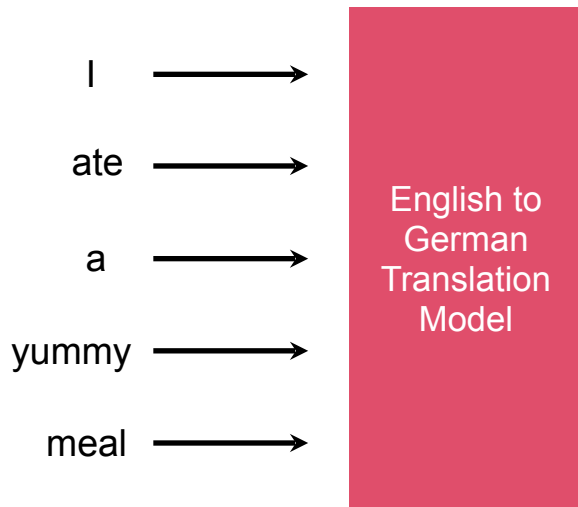


Sequence to Sequence Model



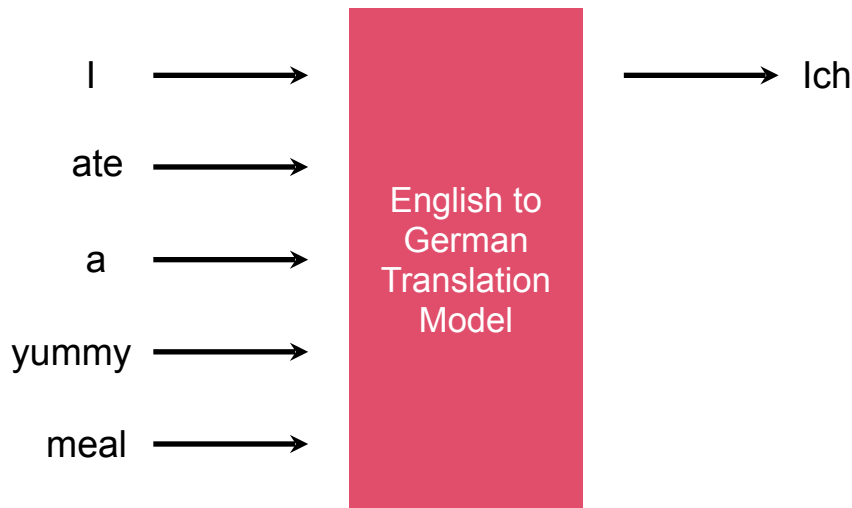


Input Sequence Fed In

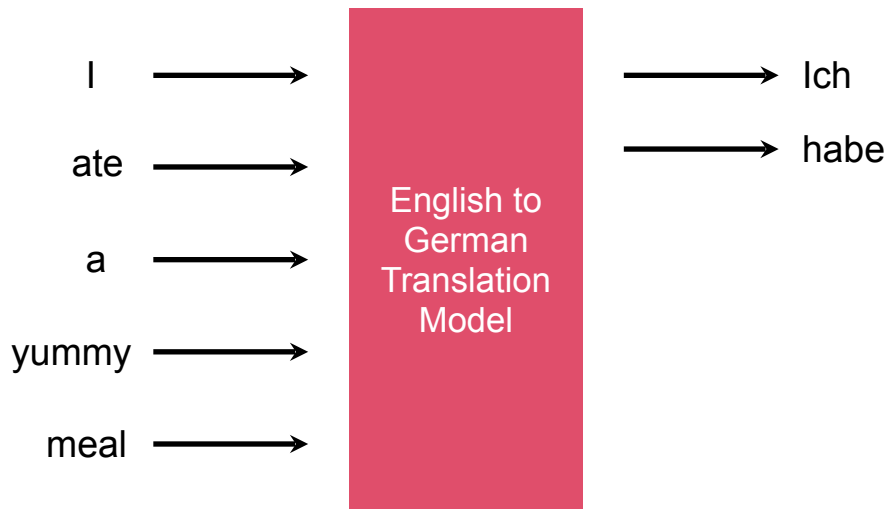




Output Sequence Produced One Word at a Time

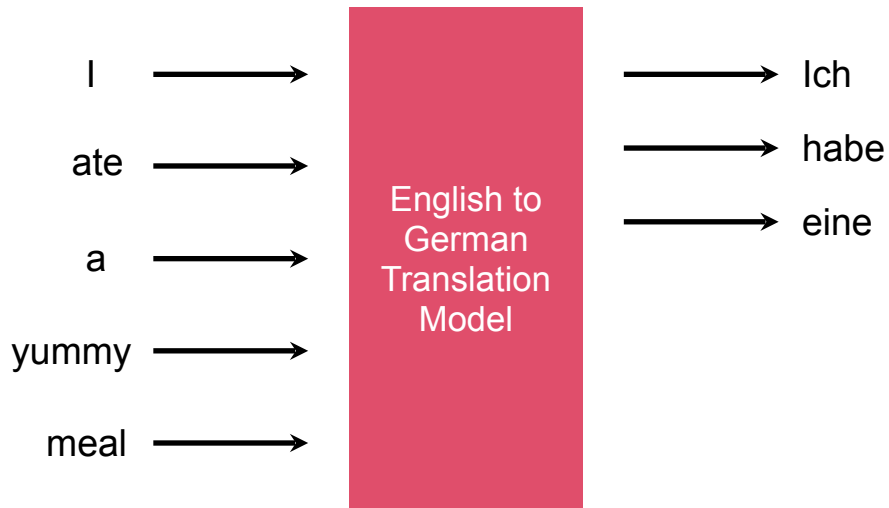


Each Output Word Used to Predict Next Word



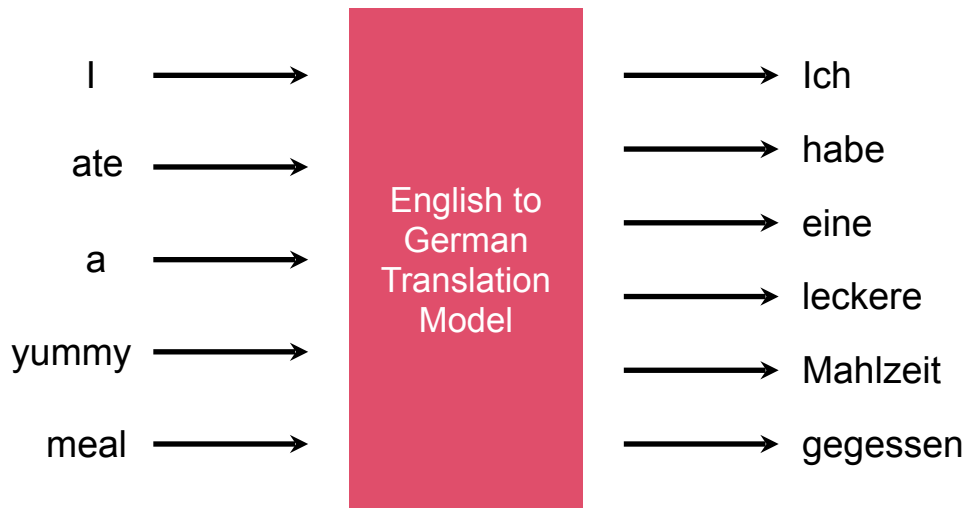
Along with the entire input sequence

Each Output Word Used to Predict Next Word



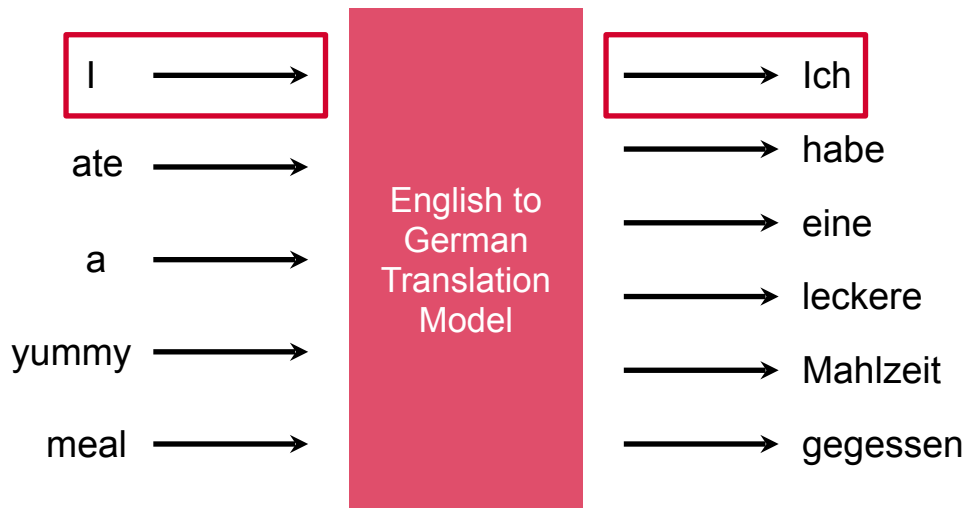


Predictions Might be Better if Attention is Employed



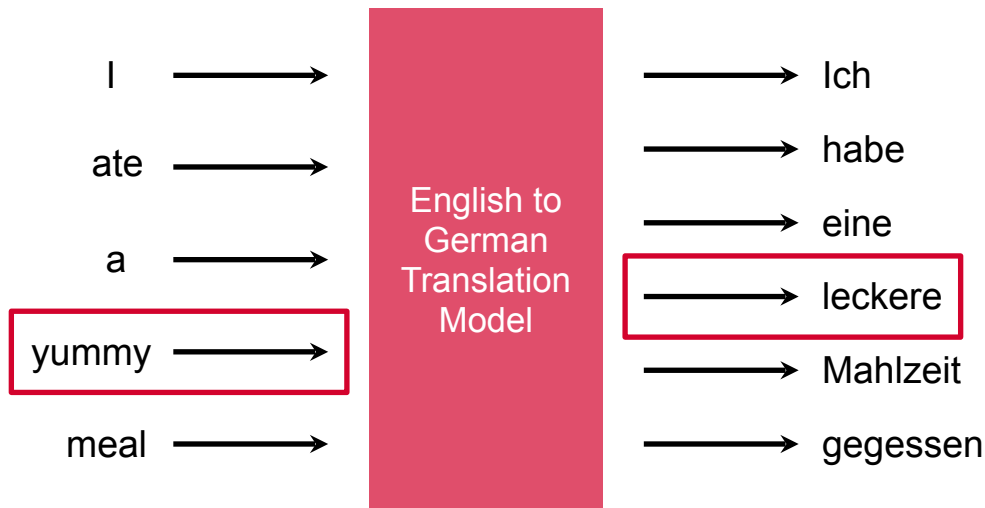


Predictions Might be Better if Attention is Employed



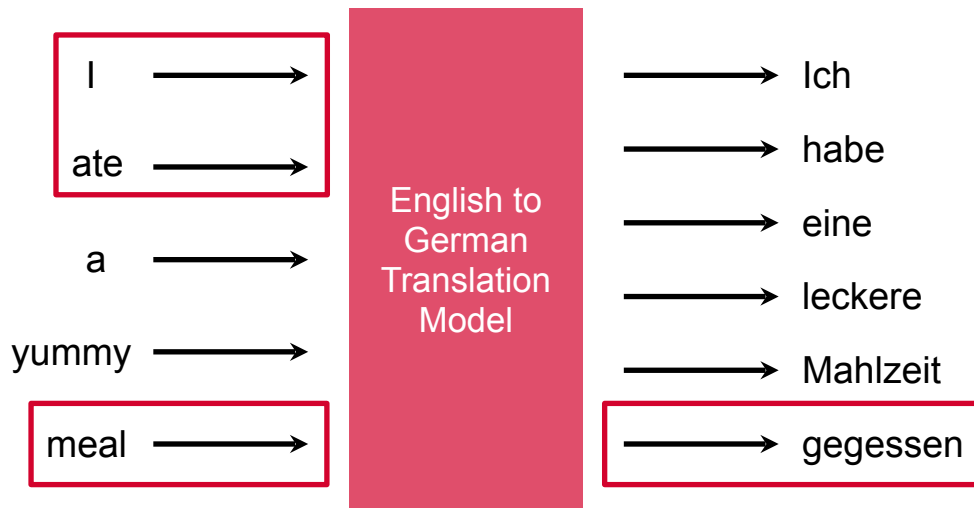


Predictions Might be Better if Attention is Employed





Predictions Might be Better if Attention is Employed





Attention

- Help models focus on the right parts of the input to generate the output
- The entire input along with attention generates better outputs
- Are the foundation of transformer models used to power the large language models of today



Large Language Models and Transformers

- The breakthrough in LLMs came with the creation of the transformer architecture
- Transformer networks are sequence-to-sequence models
- They use the concept of **attention** to focus on the right parts of the input sequence



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ChatGPT and the OpenAI Playground





ChatGPT and the OpenAI Playground

- ChatGPT – the chatbot built on top of the GPT large language model, for all users to run natural language queries
- OpenAI Playground – the web app that allows developers and researchers to work with different OpenAI models





OpenAI Playground

- A web app geared towards developers and researchers
- Hands-on access to OpenAI APIs but in a no-code manner using a nice web app
- Can access different models – even older ones – and can tune the parameters of the model in different ways



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OpenAI APIs





OpenAI APIs

OpenAI offers a set of APIs that allow developers to access and integrate various OpenAI models and capabilities into their own applications, products, and service



OpenAI APIs

- Text generation
- Image generation
- Video generation
- Text to speech
- Speech to text
- Realtime audio
- Vision
- Moderation
- Fine-tuning
- Embedding

